US Patent & Trademark Office
Patent Public Search | Text View

United States Patent Application Publication	20250284296
Kind Code	A1
Publication Date	September 11, 2025
Inventor(s)	IWAZAKI; Noritsugu et al.

VEHICLE MANUFACTURING SYSTEM AND VEHICLE MANUFACTURING METHOD

Abstract

A vehicle manufacturing system is a vehicle manufacturing system for performing control in such a way that a plurality of vehicles forming a platoon travel, the vehicle manufacturing system including: a computing unit for computing a control instruction value for controlling a speed of the vehicle; a transmitter configured to transmit the control instruction value to vehicles included in a group including two or more of the vehicles included in the platoon, the control instruction value being common among the vehicles included in the group; a speed control unit for controlling the speed of the vehicle in accordance with the control instruction value; a sensor provided to detect an inter-vehicle distance, which is a distance between the vehicles; and a group adjustment unit for changing the vehicles included in the group in accordance with the inter-vehicle distance.

Inventors: **IWAZAKI; Noritsugu (Sunto-gun, JP), KANOU; Takeshi (Seto-shi, JP), INOUE; Go (Gotenba-shi, JP), OKAMOTO; Yuki (Ebina-shi, JP), OHARA; Kento (Nissin-shi, JP), TAKAHASHI; Satoshi (Yokohama-shi, JP)**

Applicant: **TOYOTA JIDOSHA KABUSHIKI KAISHA** (Toyota-shi, JP)

Family ID: **94633634**

Assignee: **TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyota-shi, JP)**

Appl. No.: **19/041399**

Filed: **January 30, 2025**

Foreign Application Priority Data

JP	2024-035621	Mar. 08, 2024

Publication Classification

Int. Cl.: **G05D1/695** (20240101); **G05D105/45** (20240101); **G05D107/70** (20240101); **G06Q50/04** (20120101)

U.S. Cl.:

CPC **G05D1/695** (20240101); **G06Q50/04** (20130101); G05D2105/45 (20240101); G05D2107/70 (20240101)

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS
[0001] This application is based upon and claims the benefit of priority from Japanese patent application No. 2024-035621, filed on Mar. 8, 2024, the disclosure of which is incorporated herein in its entirety by reference.
BACKGROUND
[0002] The present disclosure relates to a vehicle manufacturing system and a vehicle manufacturing method.
[0003] Patent Literature 1 discloses a vehicle manufacturing system. A vehicle travels, by autonomous control or remote control, in a system for producing vehicles. [0004] [Patent Literature 1] Published Japanese Translation of PCT International Publication for Patent Application, No. 2017-538619
SUMMARY
[0005] In a vehicle manufacturing factory, a plurality of vehicles are sequentially manufactured. In a case where a plurality of vehicles perform self-propelled transportation, a server or the like transmits control instruction values to the plurality of vehicles as radio signals. If the server individually obtains, for each of the vehicles, control instruction values and transmits these control instruction values to the respective vehicles, a communication load and a processing load increase.
[0006] An object of the present disclosure is to provide a vehicle manufacturing system and a vehicle manufacturing method capable of preventing a communication load and a processing load from increasing.
[0007] A vehicle manufacturing system according to the present disclosure is a vehicle manufacturing system for performing control in such a way that a plurality of vehicles forming a platoon travel during a manufacturing process, the vehicle manufacturing system including: a computing unit for computing a control instruction value for controlling a speed of the vehicle; a transmitter configured to transmit the control instruction value to vehicles included in a group including two or more of the vehicles included in the platoon, the control instruction value being common among the vehicles included in the group; a speed control unit for controlling the speed of the vehicle in accordance with the control instruction value; a sensor provided to detect an inter-vehicle distance, which is a distance between the vehicles; and a group adjustment unit for changing the vehicles included in the group in accordance with the inter-vehicle distance.
[0008] In the above vehicle manufacturing system, an approaching vehicle whose inter-vehicle distance between it and a front vehicle has become smaller than a first threshold may be detected, and the approaching vehicle may be separated from a group including the front vehicle which is traveling ahead of the approaching vehicle.
[0009] In the above vehicle manufacturing system, a delayed vehicle whose inter-vehicle distance between it and the front vehicle has become greater than a second threshold may be detected, and the front vehicle which is traveling ahead of the delayed vehicle and the delayed vehicle may be categorized in groups different from each other.

[0010] In the above vehicle manufacturing system, in a case where it is detected that a variation in the inter-vehicle distance has become greater than a third threshold, the computing unit individually computes the control instruction values for the respective vehicles, and
[0011] In the above vehicle manufacturing system, the transmitter may transmit the control instruction values to the respective vehicles.
[0012] In the above vehicle manufacturing system, the transmitter may transmit an individual control instruction value to the vehicle that has reached a predicted part where it is expected that the inter-vehicle distance will vary.
[0013] The aforementioned vehicle manufacturing system may include a camera configured to capture an image of any one of the vehicles in the platoon, in which the computing unit may compute the control instruction value based on a result of capturing the image in the camera.
[0014] A vehicle manufacturing method according to the present disclosure is a vehicle manufacturing method for performing control in such a way that a plurality of vehicles forming a platoon travel during a manufacturing process, the vehicle manufacturing method including: computing a control instruction value for controlling a speed of the vehicle; detecting, based on a result of detection in a sensor, an inter-vehicle distance, which is a distance between one of the plurality of vehicles and another adjacent one of the plurality of vehicles; transmitting, by a transmitter, the control instruction value to vehicles included in a group including two or more of the vehicles included in the platoon, the control instruction value being common among the vehicles included in the group; controlling the speed of the vehicle in accordance with the control instruction value; and changing the vehicles included in the group in accordance with the inter-vehicle distance.
[0015] In the above vehicle manufacturing method, an approaching vehicle whose inter-vehicle distance between it and a front vehicle has become smaller than a first threshold may be detected, and the approaching vehicle may be separated from a group including the front vehicle which is traveling ahead of the approaching vehicle.
[0016] In the above vehicle manufacturing method, a delayed vehicle whose inter-vehicle distance between it and the front vehicle has become greater than a second threshold may be detected, and the front vehicle which is traveling ahead of the delayed vehicle and the delayed vehicle may be categorized in groups different from each other.
[0017] In the above vehicle manufacturing method, in a case where it is detected that a variation in the inter-vehicle distance has become greater than a third threshold, a computing unit may individually compute the control instruction values for the respective vehicles, and the transmitter may transmit the control instruction values to the respective vehicles.
[0018] In the above vehicle manufacturing method, the transmitter may transmit an individual control instruction value to the vehicle that has reached a predicted part where it is expected that the inter-vehicle distance will vary.
[0019] In the above vehicle manufacturing method, the control instruction value may be computed based on a result of capturing an image in a camera configured to capture an image of any one of the vehicles in the platoon.
[0020] According to the present disclosure, it is possible to provide a vehicle manufacturing system and a vehicle manufacturing method capable of preventing a communication load and a processing load from increasing.
[0021] The above and other objects, features and advantages of the present disclosure will become more fully understood from the detailed description given hereinbelow and the accompanying drawings.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0022] FIG. **1** is a schematic diagram showing a whole configuration of a vehicle manufacturing system according to a first embodiment;

[0023] FIG. **2** is a schematic diagram showing a part of the vehicle manufacturing system;

[0024] FIG. **3** is a block diagram showing a control system of the vehicle manufacturing system;

[0025] FIG. **4** is a schematic diagram for describing a group control example;

[0026] FIG. **5** is a schematic diagram for describing a group control example;

[0027] FIG. **6** is a schematic diagram for describing a group control example;

[0028] FIG. **7** is a flowchart showing a vehicle manufacturing method;

[0029] FIG. **8** is a schematic diagram for describing a group control example according to a second embodiment;

[0030] FIG. **9** is a flowchart for showing a vehicle manufacturing method;

[0031] FIG. **10** is a diagram for describing travel control of a vehicle;

[0032] FIG. **11** is a control block diagram for describing a travel control example 1;

[0033] FIG. **12** is a flowchart for describing the travel control example 1;

[0034] FIG. **13** is a control block diagram for describing a travel control example 2; and

[0035] FIG. **14** is a flowchart for describing the travel control example 2.

DESCRIPTION OF EMBODIMENTS

[0036] Embodiments of the present disclosure will now be described with reference to the drawings. However, the claimed disclosure is not limited to the following embodiments. Moreover, not all of the configurations described in the embodiments are essential as means for solving the problem. For the sake of clarity of the explanation, the following descriptions and drawings have been omitted and simplified as appropriate. In each drawing, the same elements have the same reference signs, and repeated descriptions have been omitted as appropriate.

First Embodiment

Vehicle Manufacturing System

[0037] With reference to FIGS. **1** and **2**, a vehicle manufacturing system **50** according to this embodiment will be described. FIG. **1** is a schematic diagram showing a configuration of the vehicle manufacturing system **50**. FIG. **2** is a diagram schematically showing two traveling vehicles **100**. FIG. **1** shows an XY orthogonal coordinate system for the sake of clarity of the explanation.

[0038] The vehicle manufacturing system (this may also be referred to as the system) **50** is used in a vehicle manufacturing factory where vehicles **100** are manufactured. The vehicle manufacturing system **50** is also used in a transport location where a transporting process such as transportation to a yard or shipment is carried out. The vehicle manufacturing system **50** may perform the following control on vehicles in the transporting process. As shown in FIG. **1**, the vehicle manufacturing system **50** includes a server **200**, a sensor **300**, and a robot **600**. The plurality of vehicles **100** are self-driving vehicles that can travel by themselves during a manufacturing process. The vehicle manufacturing system **50** performs control in such a way that the plurality of vehicles **100** perform travel in a form of a platoon.

[0039] The sensor **300** includes a communication apparatus **330** that transmits or receives data to or from the server **200**. The server **200** includes a communication apparatus **230** that transmits or receives data to or from the sensor **300**. Further, as shown in FIG. **2**, the communication apparatus **230** has a function of transmitting or receiving data to or from the vehicle **100**. Further, the vehicle **100** includes a communication apparatus **130** that receives data from the server **200**. Each of the vehicles **100** includes the communication apparatus **130**.

[0040] The communication apparatus **130**, the communication apparatus **230**, and the communication apparatus **330** may each be general-purpose equipment such as a network hub or a router apparatus. The communication apparatus **130**, the communication apparatus **230**, and the communication apparatus **330** each use, for example, general-purpose wireless communication such as WiFi (registered trademark). In each of the communication apparatus **130**, the communication apparatus **230**, and the communication apparatus **330**, an address for specifying the communication partner is set. An address for communication is, for example, an Internet Protocol (IP) address.

[0041] Each of the vehicles **100** is a vehicle whose manufacturing has not yet been completed. As shown in FIG. **1**, the vehicle **100** travels along a traveling path TR set in advance. The manufacturing of the vehicle **100** is gradually done as it travels along the traveling path TR. Specifically, while the vehicle **100** travels along the traveling path, workers (not shown), a robot **600**, and the like perform assembling of parts, switch operations, welding, inspection, and so on. The work in each manufacturing process is thus performed. The work in each manufacturing process is performed in a predetermined order, whereby the vehicle **100** is manufactured.

[0042] A plurality of vehicles **100** travel in a form of a platoon. Specifically, the vehicles **100** travel at a constant speed in such a way that distances between vehicles are constant at a predetermined distance. Further, the plurality of vehicles **100** travel at the same speed. Further, the traveling path TR includes a straight-ahead area TR**1** where a vehicle **100** travels straight ahead and a turning area TR**2** where the vehicle **100** makes a turn. In the straight-ahead area TR**1**, the traveling path TR has a linear shape.

[0043] The turning area TR**2** is a place where the vehicle **100** changes its traveling direction. In the turning area TR**2**, the vehicle **100** makes a U-turn. In the turning area TR**2**, for example, the traveling path TR has an arcuate shape having a predetermined radius of curvature. In the turning area TR**2**, the traveling path TR is a semicircle. The turning area TR**2** is provided in each end of the straight-ahead area TR**1**. For example, after traveling through the straight-ahead area TR**1** in the +X direction, the vehicle **100** reaches the turning area TR**2**. After the vehicle **100** makes a turn by 180 degrees in the turning area TR**2**, the vehicle **100** travels through the straight-ahead area TR**1** in the −X direction. On the other hand, after the vehicle **100** travels through the straight-ahead area TR**1** in the −X direction, the vehicle **100** reaches the turning area TR**2**. After the vehicle **100** makes a turn by 180 degrees in the turning area TR**2**, the vehicle **100** travels through the straight-ahead area TR**1** in the +X direction. In this manner, the manufacturing of the vehicle **100** is gradually done as it alternately passes through the straight-ahead area TR**1** and the turning area TR**2**.

[0044] Further, the turning area TR**2** is a predicted part P**2** where it is expected that inter-vehicle distances will vary. In a case where, for example, the vehicle **100** is controlled so as to decelerate at a time when the vehicle **100** makes a turn, inter-vehicle distances between this vehicle **100** and vehicles traveling before and after this vehicle **100** vary. Therefore, the turning area TR**2** is a predicted part P**2** where it is expected that inter-vehicle distances will vary.

[0045] Further, a predicted part P**1** where it is expected that the inter-vehicle distances will vary is also set in a part of the straight-ahead area TR**1** as well. The predicted part P**1** is, for example, a part where a force in the front-back direction is applied due to the presence of a slope, STOP&GO assembly points, or a manufacturing process. The slope is, for example, an uphill or a downhill. In a slope, the vehicle **100** may be accelerated or decelerated due to the gravity. Accordingly, the slope is the predicted part P**1** where it is expected that inter-vehicle distances will vary.

[0046] The STOP&GO assembly points are points where the vehicle **100** temporarily stops. For example, in a case where the robot **600** performs welding or assembling, the vehicle **100** stops. More specifically, the vehicle **100** stops when it reaches an area where the robot **600** can move. Then, in the state in which the vehicle **100** stops, the robot **600** perform an assembling operation or a welding operation. Since the vehicle **100** temporarily stops, inter-vehicle distances vary. Accordingly, each of the STOP&GO assembly points is the predicted part P**1** where it is expected that the inter-vehicle distances will vary.

[0047] The parts where a force is applied in the front-back direction in the manufacturing process are, for example, working places where a worker presses elements against a vehicle from the front or rear direction to assemble these elements. When the worker pushes elements in the front-back direction to assemble these elements, the vehicle **100** is accelerated or decelerated. The predicted part P**1** and the predicted part P**2** are associated, for example, in map information of a factory stored in the server **200**. For example, the server **200** stores map information in which coordinates indicating the predicted parts P**1** and P**2** are set. Control in the predicted parts P**1** and P**2** will be described later.

[0048] The sensor **300** is a camera that captures an image of the vehicle **100** which is moving or is stopped. The sensor **300** captures an image of one or a plurality of vehicles **100**. The sensor **300** is provided to detect inter-vehicle distances. The server **200** is able to detect the position of the vehicle **100** in the factory based on the image captured by the sensor **300**. The sensor **300** is installed, for example, on a wall surface, a support column, the ceiling or the like of the factory, and captures an image of the vehicle **100** from an oblique upward angle. The sensor **300** captures an image at an angle of view which includes two or more vehicles **100** that form a platoon. The sensor **300** may be installed at a height the same as that of the vehicles **100** and capture images of two or more vehicles **100** from a side direction thereof.

[0049] The communication apparatus **330** transmits the image captured by the sensor **300** to the server **200**. The communication apparatus **330** may transmit, besides the captured image, information obtained from the captured image to the server **200**. That is, the communication apparatus **330** transmits a result of detection detected in the sensor **300**. Note that the communication apparatus **330** may be built in the sensor **300** or may be separated from the sensor **300**. Further, the communication apparatus **330** may be shared among a plurality of sensors **300**. That is, in a case where the plurality of sensors **300** are installed, one communication apparatus **330** may transmit data to the server **200**.

[0050] As described above, after the sensor **300** captures an image of the vehicle **100**, the communication apparatus **330** transmits the captured image and the like to the server **200**. The communication apparatus **230** receives data of the captured image from the sensor **300**. The server **200** performs predetermined image processing on the image captured by the sensor **300**, whereby it is possible to specify inter-vehicle distances. For example, the server **200** calculates inter-vehicle distances D**1**-D**4**, etc. of a plurality of vehicles **100** forming a platoon. The number of vehicles forming a platoon is not particularly limited and may be any number that is equal to or greater than three.

[0051] Further, the sensor **300** for detecting inter-vehicle distances is not limited to a camera. A sensor for detecting inter-vehicle distances may be various kinds of sensors such as an RGB camera, a far-infrared camera, or a LiDAR. The sensor **300** is not limited to an optical sensor, and may instead be a radar. As a matter of course, two or more sensors **300** may be installed, or two or more sensors **300** may be used in combination with each other. For example, the sensor **300** may include a LiDAR and a camera.

[0052] The communication apparatus **330** transmits a result of the detection to the server **200**. As described above, the result of the detection transmitted from the sensor **300** may be a captured image or may be information extracted from the image. In a case where, for example, the sensor **300** has an image processing function, the sensor **300** transmits information extracted by image processing to the server **200**.

[0053] Further, as shown in FIG. **2**, the sensor **300** may be mounted on the vehicle **100**. The sensor **300** is, for example, an in-vehicle camera, a LiDAR, a radar, or the like. In a case where the sensor **300** is an in-vehicle camera, the sensor **300** captures an image of a vehicle **100** in front (this vehicle may be referred to as a front vehicle). In a case where the sensor **300** is an in-vehicle LiDAR, the sensor **300** measures a distance from the vehicle **100** to the front vehicle **100**. The communication apparatus **130** transmits the image and a result of the measurement to the server **200**.

[0054] The server **200** controls a vehicle **100** in such a way that the vehicle **100** moves along the traveling path TR. Further, the server **200** controls a plurality of vehicles **100** in such a way that the vehicles **100** perform platoon traveling. For example, the vehicles **100** travel along the traveling path TR in one line. The server **200** transmits a control signal to each vehicle **100** by the communication apparatus **230**.

[0055] Hereinafter, with reference to FIG. **3**, a control system of the vehicle manufacturing system **50** will be described. FIG. **3** is a block diagram showing the control system of the vehicle manufacturing system **50**. FIG. **3** is a schematic diagram for describing communication processing in the vehicle manufacturing system **50**. While a single vehicle **100** and a single sensor **300** are shown in FIG. **3**, a plurality of vehicles **100** and a plurality of sensors **300** are actually provided, as shown in FIG. **1**.

[0056] The server **200** includes the communication apparatus **230**, a position calculation unit **252**, a computing unit **253**, an inter-vehicle distance calculation unit **254**, a group control unit **255**, an address granting unit **256**, and an address management unit **257**. The vehicle **100** includes a vehicle control unit **115**, actuators **120**, and the communication apparatus **130**. The sensor **300** includes the communication apparatus **330**. Note that the server **200** is not limited to a physically single apparatus and a plurality of servers **200** may instead be provided in a distributed manner. For example, a database and so on may be a storage device, a cloud server, or the like that is provided separately from the processor.

[0057] While a configuration in which the position calculation unit **252**, the computing unit **253**, the inter-vehicle distance calculation unit **254**, the group control unit **255**, the address granting unit **256**, and the address management unit **257** are mounted on the server **200** will be described in the following description, the position calculation unit **252**, the computing unit **253**, the inter-vehicle distance calculation unit **254**, or the group control unit **255** may instead be mounted on the sensor **300** or the vehicle **100**. That is, the processing in the computing unit **253**, the position calculation unit **252**, the inter-vehicle distance calculation unit **254**, and the group control unit **255** may be performed in the vehicle **100** or the sensor **300**.

[0058] The position calculation unit **252** calculates position information indicating the position and the orientation of the vehicle based on the captured image. For example, the position calculation unit **252** may obtain XYZ global coordinates or an orientation in a map of the factory. At least a part of the processing in the position calculation unit **252** may be performed in the sensor **300**. For example, the sensor **300** may include a processor or the like that performs image processing. In this case, position information indicating the position and the like of the vehicle **100** is transmitted from the communication apparatus **330** to the server **200**. The position calculation unit **252** specifies each of positions of vehicles **100** on the map indicated in the map information.

[0059] The position of the vehicle **100** and the orientation of the vehicle **100** may be estimated using the captured image acquired by the sensor **300** provided in a place other than the place where the vehicle **100** is located. The position of the vehicle **100** can be acquired by calculating coordinates of measurement points of the mobile body in an image coordinate system by using, for example, the outline of the vehicle **100** detected from the captured image, and converting the calculated coordinates into coordinates in a global coordinate system. The orientation of the vehicle **100** can be estimated based on, for example, a direction of a moving vector of the mobile body calculated from changes in positions of feature points of the mobile body between frames of the captured image using an optical flow method. The orientation of the vehicle **100** may be calculated using, for example, a result of output of a yaw rate sensor or the like mounted on the vehicle **100**.

[0060] The outline of the vehicle **100** included in the captured image can be detected, for example, by inputting the captured image to the detection model that uses artificial intelligence. The detection model may be, for example, a learned machine learning model learned so as to implement one of semantic segmentation or instance segmentation. This machine learning model may be, for example, convolutional neural network (hereinafter, CNN) learned by supervised learning using a learning dataset. The learning dataset includes, for example, a plurality of training images including the mobile body, and a ground-truth label indicating whether each area in the training image is an area indicating a mobile body or an area indicating a part other than the mobile body. When CNN learning is performed, parameters of the CNN are preferably updated in such a way that the error between the result of the output by the detection model and the ground-truth label is reduced by a back propagation method.

[0061] The computing unit **253** computes control instruction values for controlling the vehicle **100**. The computing unit **253** calculates control instruction values based on the position of the vehicle **100**. The computing unit **253** sends control instructions in such a way that the vehicle **100** moves along the traveling path TR. The control instruction values here may include a speed instruction value for controlling the speed of the vehicle and a steering angle instruction value for controlling a steering angle of the vehicle. The speed instruction value may be, for example, information indicating a speed, an acceleration, or the like of the vehicle **100**. The steering angle instruction value is information indicating a steering angle or the like of the vehicle **100**. The computing unit **253** generates control instruction values regarding the movement of the vehicle **100**. In this manner, the computing unit **253** creates control instructions regarding the movement of the vehicle **100**.

[0062] The communication apparatus **230** includes a receiver **231** and a transmitter **232**. The receiver **231** receives various kinds of signals, data, or the like from the sensor **300** and the vehicle **100**. The receiver **231** receives, for example, data indicating a result of detection by the sensor **300**. Note that the data received from the sensor **300** may be image data or may be data extracted from the image data.

[0063] The transmitter **232** transmits various kinds of signals, data, or the like to the sensor **300** and the vehicle **100**. The transmitter **232** transmits, for example, control instruction values to the vehicle **100**. As a matter of course, the server **200** may transmit or receive data other than those stated above. As communication in the receiver **231** and the transmitter **232**, processing according to a general-purpose communication standard such as WiFi (registered trademark) may be used.

[0064] The communication apparatus **130** of the vehicle **100** is wireless terminal equipment for enabling wireless communication with the server **200**. An Internet Protocol (IP) address or the like is set in the communication apparatus **130**. After the communication apparatus **130** of the vehicle **100** receives control instruction values, the vehicle **100** moves in accordance with the control instruction values. The actuators **120** include a wheel motor for driving wheels, a steering motor for controlling the steering angle, a brake for stopping the vehicle, and so on. The vehicle control unit **115** generates control signals to control the actuators **120** in accordance with a control instruction. The vehicle control unit **115** may be composed of an Electronic Control Unit (ECU). Accordingly, the vehicle **100** can move along the traveling path TR.

[0065] The address granting unit **256** grants an address for communication to each vehicle **100** and the communication apparatus **130** of this vehicle **100**. For example, before the vehicle **100** starts traveling along the traveling path TR by itself, the address granting unit **256** grants an address to the communication apparatus **130**. Alternatively, an address may be associated with an ID of the vehicle **100**. Further, the address granting unit **256** grants an address to the communication apparatus **130** or the vehicle **100** in such a way that the traveling order and the address can be associated with each other. As will be described later, in a case where a platoon is not disturbed (i.e., not in disorder), the address granting unit **256** may grant a common address to the communication apparatus **130** of each of the plurality of vehicles. On the other hand, in a case where a platoon is disturbed (i.e., in disorder), the address granting unit **256** may grant individual addresses to the communication apparatuses **130** of the plurality of respective vehicles. The address granting unit **256** may grant the address before the start of production through self-propelled transportation or every time a control mode is switched.

[0066] The address management unit **257** manages addresses for communication in the entire factory. The address management unit **257** manages, for each address, whether it is being used or not being used. For example, the address management unit **257** includes a database for managing addresses. The address management unit **257** manages addresses of vehicles **100** which are carrying out self-propelled transportation as addresses that are being used. Then, for each of vehicles **100** that have already been manufactured, a flag indicating that the address is not being used is set. Accordingly, the addresses that are being used are changed to addresses that are not being used. The address management unit **257** may reuse the address that is no longer in use to control the next vehicle **100**.

[0067] The inter-vehicle distance calculation unit **254** calculates an inter-vehicle distance of vehicles **100** adjacent to each other based on a result of detection by the sensor **300**. For example, the inter-vehicle distance calculation unit **254** performs image processing on the image of the vehicle **100** captured by the sensor **300** to thereby calculate the inter-vehicle distance. The inter-vehicle distance calculation unit **254** may calculate the inter-vehicle distance based on position information. Alternatively, the inter-vehicle distance calculation unit **254** may calculate the inter-vehicle distance from a result of measurement by LiDAR or the like mounted on the vehicle **100**.

[0068] The group control unit **255** changes vehicles **100** included in one group based on the inter-vehicle distance. For example, inter-vehicle distances between each of the vehicles **100** and vehicles **100** in front and behind each of the vehicles **100** are calculated. For example, as shown in FIG. **1**, the inter-vehicle distance calculation unit **254** calculates inter-vehicle distance D**1**-D**4** of a plurality of vehicles **100** forming a platoon. A threshold or the like indicating a reference distance is set in the group control unit **255**. The group control unit **255** controls vehicles **100** included in the group based on a result of comparing the threshold with the inter-vehicle distances.

[0069] For example, the group control unit **255** stores, in advance, at least one of a first threshold, which is a lower-limit value of the reference range with respect to the inter-vehicle distance, and a second threshold, which is an upper-limit value of the reference range with respect to the inter-vehicle distance. The group control unit **255** compares the inter-vehicle distances with the thresholds. In a case where one or more of the inter-vehicle distances D**1**-D**4** has become outside of the reference range, the number of vehicles included in one group is changed. In a case where a vehicle **100** has approached to close to the front vehicle too much or the vehicle **100** has moved away from the front vehicle too much, the group control unit **255** categorizes these two vehicles **100** in groups different from each other. The group control unit **255** dynamically changes each of these groups based on a result of detection in the sensor **300**. The details of the group control unit **255** will be described later.

[0070] The server **200** may switch a group control and an individual control in a case where the server **200** transmits control instruction values to a plurality of vehicles **100**. In the group control, the server **200** transmits a common control instruction value to two or more vehicles **100** forming a group. Specifically, the plurality of vehicles **100** forming a platoon in one line belong to one group. The server **200** transmits the same control instruction value to two or more vehicles **100** that belong to one group. In the individual control, individual control instruction values are transmitted to the respective vehicles **100**.

[0071] After the communication apparatus **130** receives the control instruction value, the vehicle control unit **115** generates a control signal for controlling the actuators **120** in accordance with the control instruction value. For example, the vehicle control unit **115** controls a wheel motor or the like in such a way that the speed or the acceleration indicated by the control instruction value is achieved.

[0072] For example, in the straight-ahead area TR**1**, the computing unit **253** creates a common control instruction value for the plurality of vehicles **100** in the platoon. In a case where a group control is performed, the computing unit **253** creates the same speed instruction value for a plurality of vehicles **100** aligned in one line. Then, the transmitter **232** transmits the common control instruction value to the plurality of vehicles **100**. Therefore, the plurality of vehicles **100** that belong to one group can travel at the same speed.

[0073] Accordingly, it is possible to prevent the computing load and the communication load in the server **200** from increasing. For example, a control instruction value such as the speed or the acceleration is commonly used among a plurality of vehicles **100**. Accordingly, in the server **200**, it is not necessary to perform processing for obtaining the control instruction value for each of the vehicles **100**. For example, the computing unit **253** obtains a control instruction value for the leading vehicle **100** in the platoon. Then, the transmitter **232** transmits this control instruction value to the leading vehicle **100** and the following vehicles **100** through multicast. The transmitter **232** is able to transmit the control instruction value to the plurality of vehicles **100** through multicast. Accordingly, it is possible to prevent the communication load from increasing.

[0074] Hereinafter, with reference to FIGS. **4-6**, examples of the group control in the group control unit **255** will be described. FIG. **4** and so on show seven vehicles **100***a*-**100***g*. The seven vehicles **100***a*-**100***g* travel straight ahead along a traveling path TR. The vehicles **100***a*-**100***g* travel in a platoon in one line. Further, an inter-vehicle distance between the first vehicle **100***a* and the second vehicle **100***b* is referred to as an inter-vehicle distance D**1**. An inter-vehicle distance between the second vehicle **100***b* and the third vehicle **100***c* is referred to as an inter-vehicle distance D**2**. Likewise, the following inter-vehicle distances are referred to as inter-vehicle distance D**3**-D**6**.

Group Control Example 1

[0075] With reference to FIG. **4**, a group control example 1 in the group control unit **255** will be described. A first threshold TH**1** is set in the group control unit **255**. The group control unit **255** compares each of the inter-vehicle distances D**1**-D**6** with the first threshold TH**1**.

[0076] FIG. **4** is a schematic diagram for describing processing in a case where a vehicle **100** has approached to close to the front vehicle. It is assumed here that the inter-vehicle distance D**3** has become smaller than the first threshold TH**1**. The group control unit **255** detects that the fourth vehicle **100***d* is an approaching vehicle whose inter-vehicle distance D**3** between it and the front vehicle has become smaller than the first threshold TH**1**. The vehicle **100***c* just in front of the vehicle **100***d* is the front vehicle **100***c*. The group control unit **255** causes the approaching vehicle **100***d* to be separated from a group including the front vehicle **100***c*. Specifically, the group control unit **255** classifies the vehicles **100***a*-**100***c* in a first group G**1** and classifies the vehicles **100***d*-**100***g* in a second group G**2**.

[0077] The computing unit **253** computes different control instruction values for the first group G**1** and the second group G**2**, respectively. For example, the computing unit **253** computes the control instruction values in such a way that the vehicles **100***d*-**100***g* included in the second group G**2** move slower than the vehicles **100***a*-**100***c* included in the first group G**1**. That is, the vehicles **100***a*-**100***c* travel forward at a speed higher than that of the vehicles **100***d*-**100***g*. According to the above configuration, the vehicles **100***d*-**100***g* included in the second group G**2** are gradually moving away from the vehicles **100***a*-**100***c* included in the first group G**1**. Then, as shown in FIG. **5**, when the inter-vehicle distance D**3** has become greater than the first threshold TH**1**, the group control unit **255** classifies the vehicles **100***a*-**100***g* in one group G.

[0078] As described above, in a case where an inter-vehicle distance has become too short, the group control unit **255** adjusts vehicles that belong to a group. Accordingly, the approaching vehicle **100***d* that has approached to close to the front vehicle **100***c* too much can be made to move away from the front vehicle **100***c*, whereby it is possible to avoid a contact or the like between the vehicles.

Group Control Example 2

[0079] With reference to FIG. **6**, a group control example 2 in the group control unit **255** will be described. FIG. **6** is a schematic diagram for describing processing in a case where a vehicle moves away from the front vehicle. A second threshold TH**2** is set in the group control unit **255**. The second threshold TH**2** is a value greater than the first threshold TH**1**.

[0080] It is assumed here that the inter-vehicle distance D**4** is greater than the second threshold TH**2**. The group control unit **255** detects that the fifth vehicle **100***e* is a delayed vehicle whose inter-vehicle distance D**4** between it and a vehicle in front of it has become greater than the second threshold TH**2**. The vehicle **100***d* just in front of the delayed vehicle **100***e* will be referred to as a front vehicle. The group control unit **255** causes the delayed vehicle **100***e* to be separated from a group including the front vehicle **100***d*. Specifically, the group control unit **255** classifies the vehicles **100***a*-**100***d* in a first group G**1** and classifies the vehicles **100***e*-**100***g* in a second group G**2**.

[0081] The computing unit **253** computes control instruction values different from each other for the first group G**1** and the second group G**2**, respectively. For example, the computing unit **253** calculates the control instruction values in such a way that the vehicles **100***e*-**100***g* in the second group G**2** move faster than the vehicles **100***a*-**100***d* in the first group G**1**. That is, the vehicles **100***a*-**100***d* travel forward at a speed lower than that of the vehicles **100***e*-**100***g*. According to the above configuration, the vehicles **100***e*-**100***g* in the second group G**2** gradually approach the vehicles **100***a*-**100***d* in the first group G**1**. Then, as shown in FIG. **5**, when the inter-vehicle distance Dd has become greater than the second threshold TH**2**, the group control unit **255** classifies the vehicles **100***a*-**100***g* in one group G.

[0082] As described above, in a case where an inter-vehicle distance has become too great, the group control unit **255** adjusts vehicles that belong to a group. Accordingly, the delayed vehicle **100***e* can be made to move closer to the front vehicle **100***d*, whereby it is possible to efficiently manufacture vehicles.

[0083] According to the above configuration, even in a case where a platoon is disturbed, it is possible to resolve disturbance of the platoon. That is, a plurality of vehicles **100** perform platoon traveling at predetermined intervals. Accordingly, it is possible to manufacture the vehicles **100** more safely and more efficiently, whereby productivity may be improved.

[0084] FIG. **7** is a flowchart showing a vehicle manufacturing method according to this embodiment. Referring to FIG. **7**, the vehicle manufacturing method will be described. As described above, it is assumed that a plurality of vehicles **100** forming a platoon travel along a traveling path TR.

[0085] The sensor **300** detects a vehicle **100** (S**11**). In a case where the sensor **300** is a camera, the sensor **300** captures an image of the vehicle **100**. Next, the inter-vehicle distance calculation unit **254** calculates inter-vehicle distances based on a result of detection by the sensor **300** (S**12**). In this example, the inter-vehicle distance calculation unit **254** performs image processing or the like, to thereby calculate inter-vehicle distances between a plurality of vehicles **100** forming a platoon.

[0086] Next, the group control unit **255** determines whether or not the inter-vehicle distance is smaller than a first threshold TH**1** (S**13**). In a case where the inter-vehicle distance is not smaller than the first threshold TH**1** (NO in S**13**), the group control unit **255** determines whether or not the inter-vehicle distance is greater than the second threshold TH**2** (S**14**).

[0087] In a case where the inter-vehicle distance is greater than the first threshold TH**1** (YES in S**13**) or in a case where the inter-vehicle distance is greater than the second threshold TH**2** (YES in S**14**), the group control unit **255** changes vehicles that belong to the group (S**15**). When, for example, the inter-vehicle distance is smaller than the first threshold TH**1**, the group control unit **255** separates groups with a boundary between the front vehicle and the approaching vehicle. In a case where the inter-vehicle distance is greater than the second threshold TH**2**, the group control unit **255** separates groups with a boundary between the front vehicle and the delayed vehicle.

[0088] The computing unit **253** computes a control instruction value for each of the groups (S**16**). For example, the computing unit **253** obtains a control instruction value for vehicles **100** in the first group and a control instruction value for vehicles **100** in the second group. The transmitter **232** transmits the control instruction values to the respective vehicles **100** (S**17**). The vehicle control unit **115** performs control in such a way that each vehicle **100** travels at the speed or the acceleration indicated by the common control instruction value. Accordingly, the vehicles **100** in the first group and the vehicles **100** in the second group travel at speeds different from each other.

[0089] In a case where the inter-vehicle distance is not greater than the second threshold TH**2** (NO in S**14**), the computing unit **253** computes a control instruction value (S**18**). In this manner, in a case where the inter-vehicle distance is within an appropriate range, the computing unit **253** computes a common control instruction value for vehicles **100** that belong to one group. The computing unit **253** computes a common control instruction value for a plurality of vehicles **100**. It is therefore possible to prevent a processing load from increasing. The transmitter **232** transmits the control instruction value to each of the vehicles **100** (S**19**). The vehicle control unit **115** performs control in such a way that the plurality of vehicles **100** travel at the speed or the acceleration indicated by the common control instruction value. It is therefore possible to prevent a communication load from increasing.

Second Embodiment

[0090] In a second embodiment, in a case where a variation in the inter-vehicle distances is great, the server **200** changes vehicles that belong to a group. Hereinafter, with reference to FIG. **8**, an operation according to the second embodiment will be described. Since the basic configuration and processing are similar to those in the first embodiment, the descriptions thereof will be omitted. Further, the second embodiment may be performed along with the first embodiment or may be independently performed.

Group Control Example 3

[0091] FIG. **8** is a schematic diagram for describing an operation in a group control example 3 in the group control unit **255**. In FIG. **8**, seven vehicles **100***a*-**100***g* travel, like in FIG. **4**, etc. FIG. **8** shows a platoon in a case where a variation in the inter-vehicle distances has become great. A third threshold TH**3** is set in the group control unit **255**. The third threshold TH**3** is a value for determining whether or not the variation in the inter-vehicle distances is great. The variation in the inter-vehicle distances is indicated, for example, by a standard deviation of inter-vehicle distances D**1**-D**6**. It is further assumed that the inter-vehicle distances D**1**-D**6** are values, each of which falls between the first threshold TH**1** and the second threshold TH**2**.

[0092] The group control unit **255** obtains a standard deviation of the plurality of inter-vehicle distances D**1**-D**6**. The group control unit **255** compares the standard deviation with the third threshold TH**3**. In a case where it is detected that the standard deviation is greater than the third threshold TH**3**, the group control unit **255** changes vehicles that belong to the group. Specifically, the group control unit **255** switches the control mode to an individual control. In the individual control, the computing unit **253** computes individual control instruction values for the respective vehicles **100**. The computing unit **253** computes control instruction values in accordance with the inter-vehicle distances for the vehicles **100***a*-**100***g*. That is, the vehicles **100***a*-**100***g* travel forward at speeds different from one another.

[0093] The computing unit **253** computes a speed control value in such a way that an inter-vehicle distance between a vehicle and the front vehicle approaches a predetermined value. That is, in a case where the inter-vehicle distance to the front vehicle is relatively large, the rear vehicle is made to travel at a speed faster than that of the front vehicle so that the rear vehicle approaches the front vehicle. In a case where the inter-vehicle distance between a vehicle and the front vehicle is relatively small, the rear vehicle is made to travel at a speed lower than that of the front vehicle so that the rear vehicle moves away from the front vehicle. According to the above configuration, the variation in the inter-vehicle distances is reduced. Therefore, when the variation in the inter-vehicle distances becomes smaller than the third threshold TH**3**, as shown in FIG. **5**, the vehicles **100***a*-**100***g* are classified in one group G.

[0094] In the group control, the server **200** may manage a common address as addresses of the plurality of vehicles **100**. It is assumed, for example, that an IP address is assigned to radio equipment of each of the vehicles **100** as a destination address. It is further assumed that the server **200** can rewrite the IP addresses. That is, the server **200** manages IP addresses in the factory on a database. Then, the server **200** manages whether each IP address is being used or unused.

[0095] In the group control, the address granting unit **256** grants a common IP address to the plurality of vehicles **100**. The same IP address is assigned to the plurality of vehicles **100**, whereby the server **200** is able to easily perform a group control of the plurality of vehicles **100**. In the group control, the address granting unit **256** grants a common address to a plurality of vehicles **100** which belong to one group. The transmitter **232** transmits a common control instruction value to one IP address, which is a destination address, through multicast.

[0096] In a case where the group control and the individual control are switched, the server **200** transmits a signal for rewriting the IP address to the vehicle **100**. When the control is switched to the group control, the IP address is rewritten in such a way that the communication apparatuses **130** of the respective vehicles **100** have a common address. When the control is switched to the individual control, the IP address is rewritten in such a way that the communication apparatus **130** of the vehicle **100** have an individual address. Then, the communication apparatus **130** transmits, to the server **200**, a signal indicating that the IP address has been rewritten. Upon receiving the signal from the communication apparatus **130**, the server **200** rewrites the data in the database so as to manage the IP address.

[0097] FIG. **9** is a flowchart showing a vehicle manufacturing method according to this embodiment. With reference to FIG. **9**, the vehicle manufacturing method will be described. It is assumed, as described above, that a plurality of vehicles **100** forming a platoon are traveling along a traveling path TR. First, the server **200** sets in advance, for each of the vehicles **100**, an IP address for unicast and an IP address for multicast (S**21**). Accordingly, two addresses are set in a communication apparatus **130** of one vehicle **100**. The IP address for multicast is common among the plurality of vehicles **100**. The IP address for unicast is individually set in each of the plurality of vehicles **100**.

[0098] The sensor **300** detects a vehicle **100** (S**22**). In a case where the sensor **300** is a camera, the sensor **300** captures an image of the vehicle **100**. Next, the inter-vehicle distance calculation unit **254** calculates inter-vehicle distances based on a result of detection in the sensor **300** (S**23**). In this example, the inter-vehicle distance calculation unit **254** performs image processing or the like to thereby calculate inter-vehicle distances between a plurality of vehicles **100** forming a platoon.

[0099] Next, the group control unit **255** determines whether or not a variation in the inter-vehicle distances is smaller than a third threshold TH**3** (S**24**). When the variation in the inter-vehicle distances is not greater than the third threshold TH**3** (NO in S**24**), the computing unit **253** computes a common control instruction value for the plurality of vehicles **100** (S**25**). It is therefore possible to prevent a processing load from increasing. The transmitter **232** transmits the control instruction value to the IP address for multicast (S**26**). A common IP address for multicast is set in the vehicles **100** that belong to the group. The transmitter **232** performs multicast transmission, whereby it is possible to prevent a communication load from increasing.

[0100] In a case where the variation in the inter-vehicle distances is greater than the third threshold TH**3** (YES in S**24**), the computing unit **253** computes individual control instruction values for a plurality of respective vehicles **100** (S**27**). The transmitter **232** transmits the control instruction values to the IP addresses for unicast (S**28**). Accordingly, it is possible to transmit appropriate control instruction values to the respective vehicles. Accordingly, it is possible to prevent inter-vehicle distances from varying.

[0101] Further, in the predicted parts P**1** and P**2** where it is expected that the inter-vehicle distances will vary, the server **200** may switch the control mode of vehicles **100** to the individual control. For example, at a timing when vehicles **100** enter the predicted parts P**1** and P**2**, the server **200** switches the control mode of these vehicles **100** to the individual control. That is, the server **200** excludes vehicles **100** at the predicted parts P**1** and P**2** from a group of the group control. In the predicted parts P**1** and P**2**, it is predicted that there will be a disturbance in the inter-vehicle distance. Therefore, the server **200** switches the control mode of the vehicles to the individual control. Accordingly, it is possible to prevent the platoon from being disturbed in advance. Further, at a timing when vehicles **100** pass through the predicted parts P**1** and P**2**, the control mode of these vehicles **100** may be switched to the group control again. At a timing when the vehicles **100** have passed through the predicted parts P**1** and P**2**, the server **200** performs processing for allowing

these vehicles **100** which have been subjected to individual control to be grouped again.
[0102] In the predicted parts P**1** and P**2**, the server **200** may switch the control mode to the individual control. The server **200** may perform control. That is, the transmitter **232** may transmit individual control instruction values to respective vehicles **100** that have reached the predicted parts P**1** and P**2** where it is expected that the inter-vehicle distances will vary. At a timing when the vehicles **100** enter the predicted parts P**1** and P**2**, the server **200** switches the group control to the individual control. In the predicted parts P**1** and P**2**, the server **200** transmits the individual control instruction values to the respective vehicles **100**. The vehicles **100** at the predicted parts P**1** and P**2** travel at their individual vehicle speeds. Therefore, each of the vehicles **100** may travel at an appropriate speed.
[0103] Further, at a timing when a vehicle **100** exits the predicted parts P**1** and P**2**, the server **200** switches the individual control to the group control. Accordingly, the control mode of vehicles **100** other than those in the predicted parts P**1** and P**2** is switched to the group control. The server **200** transmits the common control instruction value to a plurality of vehicles **100** traveling in parts other than the predicted parts P**1** and P**2**. Accordingly, it is possible to prevent the communication load from increasing.
[0104] While an example in which the control mode of controlling the speed instruction value regarding the speed is switched from the group control to the individual control has been described in the aforementioned description, a control mode of controlling the steering angle instruction value regarding the steering angle may instead be switched from the group control to the individual control. For example, in the straight-ahead area TR**1**, the server **200** transmits a common steering angle to a plurality of vehicles **100** which belong to one group as the steering angle instruction value.
[0105] Further, the sensor **300** may include a camera that captures an image of one of vehicles in a platoon. Accordingly, it is possible to appropriately detect each of the inter-vehicle distances D**1**-D**3** of the plurality of vehicles **100**. Then, the computing unit **253** computes a control instruction value based on a result of capturing images in a camera. For example, the computing unit **253** calculates, by image processing, the position of the vehicle **100** and the inter-vehicle distances. The computing unit **253** obtains a control instruction value for the leading vehicle **100**. Then, in the group control, a common control instruction value is transmitted to the vehicles **100** in the group including the leading vehicle **100**. Accordingly, it is possible to prevent the communication load from increasing and to cause the vehicles **100** in the platoon to travel with a high accuracy. For example, the sensor **300** may capture an image of the leading vehicle in the platoon. As a matter of course, the sensor **300** may capture images of two or more vehicles **100**.
[0106] Hereinafter, travel control examples for controlling traveling of the vehicle **100** in a system will be explained.
A. Travel Control Example 1
[0107] FIG. **10** is a conceptual diagram showing a configuration of a system **50** according to a travel control example 1. The system **50** includes a plurality of vehicles **100**, each of which corresponds to a mobile body, a server **200**, and one or more sensors **300**.
[0108] Note that, when the mobile body is other than a vehicle, the terms “vehicle” and “car” in the present disclosure can be replaced by a “mobile body” as appropriate, and the term “travel” can be replaced by “move” as appropriate.
[0109] The vehicle **100** is configured to be able to travel by unmanned driving. The “unmanned driving” means driving not dependent on a driver's traveling operation. The traveling operation means an operation regarding at least one of “run”, “turn”, or “stop” of the vehicle **100**. The unmanned driving is achieved by automatic or manual remote control that uses an apparatus located in the outside of the vehicle **100**, or by autonomous control of the vehicle **100**. Any passenger who does not perform the traveling operation may get on the vehicle **100** traveled by unmanned driving. The passenger who does not perform the traveling operation includes, for

example, a person who is just sitting on a seat of the vehicle **100**, and a person who is performing work such as an operation of assembling, an operation of inspection, or an operation of switches, which is an operation other than the traveling operation, while getting in the vehicle **100**. Note that the driving by the traveling operation of the driver may be referred to as "manned driving".

[0110] The "remote control" here includes "complete remote control" in which all the operations of the vehicle **100** are completely determined from the outside of the vehicle **100** and "partial remote control" in which a part of the operations of the vehicle **100** is determined from the outside of the vehicle **100**. Further, "autonomous control" includes "complete autonomous control" in which the vehicle **100** autonomously controls its own operation without receiving any piece of information from an external apparatus outside the vehicle **100** and "partial autonomous control" in which the vehicle **100** autonomously controls its own operation using information received from the external apparatus outside the vehicle **100**.

[0111] In this embodiment, the system **50** is used in a factory FC which manufactures the vehicles **100**. The reference coordinate system of the factory FC is a global coordinate system GC. That is, a desired position in the factory FC is expressed by coordinates of X, Y, and Z in the global coordinate system GC. The factory FC includes a first place PL**1** and a second place PL**2**. The first place PL**1** and the second place PL**2** are connected to each other by a traveling path TR along which the vehicle **100** can travel. A plurality of sensors **300** are installed along the traveling path TR in the factory FC. The positions of the respective sensors **300** in the factory FC are adjusted in advance. The vehicle **100** moves from the first place PL**1** to the second place PL**2** along the traveling path TR by unmanned driving.

[0112] FIG. **11** is a block diagram showing a configuration of the system **50**. The vehicle **100** includes a vehicle control apparatus **110** for controlling each part of the vehicle **100**, actuators **120** including one or more actuators that drive under a control of the vehicle control apparatus **110**, and a communication apparatus **130** for communicating with an external apparatus such as the server **200** by wireless communication. The actuators **120** include an actuator of a drive apparatus for accelerating the vehicle **100**, an actuator of a steering apparatus for changing a traveling direction of the vehicle **100**, and an actuator of a control apparatus for decelerating the vehicle **100**.

[0113] The vehicle control apparatus **110** is composed of a computer including a processor **111**, a memory **112**, an input/output interface **113**, and an internal bus **114**. The processor **111**, the memory **112**, and the input/output interface **113** are connected to one another via the internal bus **114** in such a way that they can communicate with one another. The actuators **120** and the communication apparatus **130** are connected to the input/output interface **113**. The processor **111** executes a program PG**1** stored in the memory **112**, thereby implementing various functions including a function as a vehicle control unit **115**.

[0114] The vehicle control unit **115** causes the vehicle **100** to travel by controlling the actuators **120**. The vehicle control unit **115** is able to cause the vehicle **100** to travel by controlling the actuators **120** using a travel control signal received from the server **200**. The travel control signal is a control signal for traveling the vehicle **100**. In this embodiment, the travel control signal includes an acceleration and a steering angle of the vehicle **100** as parameters. In another embodiment, the travel control signal may include, in place of or in addition to the acceleration of the vehicle **100**, a speed of the vehicle **100** as a parameter.

[0115] The server **200** is composed of a computer including a processor **201**, a memory **202**, an input/output interface **203**, and an internal bus **204**. The processor **201**, the memory **202**, and the input/output interface **203** are connected to one another via the internal bus **204** in such a way that they can communicate with one another. A communication apparatus **205** for communicating with various kinds of apparatuses provided outside the server **200** is connected to the input/output interface **203**. The communication apparatus **205** can communicate with the vehicle **100** by wireless communication and can communicate with each of the sensors **300** by wired communication or wireless communication. The processor **201** executes a program PG**2** stored in the memory **202**, thereby implementing various functions including the function as a remote control unit **210**.

[0116] The remote control unit **210** acquires a result of detection by the sensors, generates a travel control signal for controlling the actuators **120** of the vehicle **100** using the result of the detection, and transmits the travel control signal to the vehicle **100**, thereby causing the vehicle **100** to travel by remote control. The remote control unit **210** may generate not only the travel control signal but also, for example, control signals for controlling actuators for operating various kinds of auxiliary equipment provided in the vehicle **100** or various kinds of equipment such as a windshield wiper, power windows, or lamps. That is, the remote control unit **210** may operate these various kinds of equipment or various kinds of auxiliary equipment by remote control.

[0117] The sensor **300** is a sensor that is provided outside the vehicle **100**. The sensor **300** according to this embodiment is a sensor that captures the vehicle **100** from the outside of the vehicle **100**. The sensor **300** includes a communication apparatus (not shown) and can communicate with other apparatuses such as the server **200** and so on by wired communication or wireless communication.

[0118] Specifically, the sensor **300** is composed of a camera. The camera as the sensor **300** captures an image including the vehicle **100**, and outputs the captured image as a result of detection.

[0119] FIG. **12** is a flowchart showing a processing procedure of travel control of the vehicle **100** according to the travel control example. In the processing procedure shown in FIG. **12**, the processor **201** of the server **200** functions as the remote control unit **210** by executing the program PG**2**. Further, the processor **111** of the vehicle **100** functions as the vehicle control unit **115** by executing the program PG**1**.

[0120] In Step S**110**, the processor **201** of the server **200** acquires vehicle position information of the vehicle **100** using the result of the detection output from the sensor **300**. The vehicle position information is position information that forms a basis for generating a travel control signal. In this embodiment, the vehicle position information includes the position and the orientation of the vehicle **100** in a global coordinate system GC of the factory FC. Specifically, in Step S**110**, the processor **201** acquires the vehicle position information using the captured image acquired from the camera, which is the sensor **300**.

[0121] Specifically, in Step S**110**, the processor **201** detects, for example, the outline of the vehicle **100** from the captured image, calculates the coordinate system of the captured image, that is, coordinates of measurement points of the vehicle **100** in the local coordinate system, and converts the calculated coordinates into coordinates in a global coordinate system GC, thereby acquiring the position of the vehicle **100**. The outline of the vehicle **100** included in the captured image can be detected, for example, by inputting the captured image to the detection model DM that uses artificial intelligence. The detection model DM is prepared, for example, in the system **50** or in the outside of the system **50** and is stored in the memory **202** of the server **200** in advance. The detection model DM may be, for example, a learned machine learning model learned so as to implement one of semantic segmentation or instance segmentation. This machine learning model may be, for example, convolutional neural network (hereinafter, CNN) learned by supervised learning using a learning dataset. The learning dataset includes, for example, a plurality of training images including the vehicle **100**, and a label indicating whether each area in the training image is an area indicating the vehicle **100** or an area indicating a part other than the vehicle **100**. When CNN learning is performed, parameters of the CNN are preferably updated in such a way that the error between the result of the output by the detection model DM and the label is reduced by a back propagation method. Further, the processor **201** is able to acquire the orientation of the vehicle **100** by estimating it based on the direction of the moving vector of the vehicle **100** calculated from changes in positions of feature points of the vehicle **100** between frames of the captured image using an optical flow method.

[0122] In Step S**120**, the processor **201** of the server **200** determines the target position that the vehicle **100** should go next. In this embodiment, the target position is expressed by coordinates of X, Y, and Z in a global coordinate system GC. The memory **202** of the server **200** stores a reference route RR, which is a route along which the vehicle **100** should travel, in advance. The route is expressed by a node indicating the departure place, nodes indicating passage points, a node that indicates the target position, and a link connecting the respective nodes. The processor **201** determines the target position that the vehicle **100** should go next using the vehicle position information and the reference route RR. The processor **201** determines the target position on the reference route RR which is ahead of the current position of the vehicle **100**.

[0123] In Step S**130**, the processor **201** of the server **200** generates a travel control signal for traveling the vehicle **100** toward the determined target position. The processor **201** calculates the traveling speed of the vehicle **100** from the transition of the position of the vehicle **100** and compares the calculated traveling speed with the target speed. In general, when the traveling speed is lower than the target speed, the processor **201** determines the acceleration in such a way that the vehicle **100** accelerates. On the other hand, when the traveling speed is higher than the target speed, the processor **201** determines the acceleration in such a way that the vehicle **100** decelerates. Further, when the vehicle **100** is positioned on the reference route RR, the processor **201** determines the steering angle and the acceleration to prevent the vehicle **100** from being deviated from the reference route RR. On the other hand, when the vehicle **100** is not positioned on the reference route RR, that is, when the vehicle **100** is deviated from the reference route RR, the processor **201** determines the steering angle and the acceleration in such a way that the vehicle **100** returns onto the reference route RR.

[0124] In Step S**140**, the processor **201** of the server **200** transmits a generated travel control signal to the vehicle **100**. The processor **201** repeats, in a predetermined cycle, acquisition of the position of the vehicle **100**, determination of the target position, generation of the travel control signal, transmission of the travel control signal, and the like.

[0125] In Step S**150**, the processor **111** of the vehicle **100** receives the travel control signal transmitted from the server **200**. In Step S**160**, the processor **111** of the vehicle **100** controls the actuators **120** using the received travel control signal, thereby causing the vehicle **100** to travel at an acceleration and a steering angle indicated in the travel control signal. The processor **111** repeats reception of the travel control signal, and control of the actuators **120** in a predetermined cycle. With the system **50** in this example, it is possible to cause the vehicle **100** to travel by remote control and to move the vehicle **100** without using transport equipment such as cranes or conveyors.

B: Travel Control Example 2

[0126] FIG. **13** is an explanatory diagram showing a schematic configuration of a system **50***v* according to a travel control example 2. In this example, the system **50***v* is different from that in the travel control example 1 in that the system **50***v* does not include the server **200**. Further, a vehicle **100***v* in this configuration can travel by autonomous control of the vehicle **100***v*. The other configurations are the same as those stated above unless otherwise specified.

[0127] In this example, a processor **111***v* of a vehicle control apparatus **110***v* functions as a vehicle control unit **115***v* by executing a program PG**1** stored in a memory **112***v*. The vehicle control unit **115***v* acquires a result of output by a sensor, generates a travel control signal using the result of the output, and outputs the generated travel control signal to operate the actuators **120**, thereby enabling the vehicle **100***v* to travel by autonomous control. In this example, the memory **112***v* stores, besides the program PG**1**, a detection model DM and a reference route RR in advance.

[0128] FIG. **14** is a flowchart showing a processing procedure of a travel control of the vehicle **100***v* in Travel control Example 2. In the processing procedure shown in FIG. **14**, the processor **111***v* of the vehicle **100***v* functions as the vehicle control unit **115***v* by executing the program PG**1**.

[0129] In Step S**210**, the processor **111***v* of the vehicle control apparatus **110***v* acquires vehicle position information using a result of detection output from a camera, which is the sensor **300**. In

Step S**220**, the processor **111***v* determines the target position that the vehicle **100***v* should go next. In Step S**230**, the processor **111***v* generates a travel control signal for causing the vehicle **100***v* to travel toward the determined target position. In Step S**240**, the processor **111***v* controls the actuators **120** using the generated travel control signal, thereby causing the vehicle **100***v* to travel according to a parameter indicated in the travel control signal. The processor **111***v* repeats acquisition of the vehicle position information, determination of the target position, generation of the travel control signal, and control of the actuators in a predetermined cycle. With the system **50***v* in this example, it is possible to cause the vehicle **100***v* to travel by autonomous control of the vehicle **100***v* without remotely controlling the vehicle **100***v* by the server **200**.

YY: Other Travel Control Examples

[0130] (YY1) In the above examples, the sensor **300** is a camera. On the other hand, the sensor **300** may not be a camera, and may instead be, for example, Light Detection And Ranging (LiDAR). In this case, the result of the detection output from the sensor **300** may be three dimensional point cloud data indicating the vehicle **100**. In this case, the server **200** and the vehicle **100** may acquire the vehicle position information by template matching that uses three dimensional point cloud data indicating the result of the detection and reference point cloud data that is prepared in advance.

[0131] (YY2) In the travel control example 1, the server **200** executes processing from the acquisition of the vehicle position information to the generation of the travel control signal. On the other hand, the vehicle **100** may perform at least a part of the processing from the acquisition of the vehicle position information to the generation of the travel control signal. For example, the following forms from (1) to (3) may be employed.

[0132] (1) The server **200** may acquire vehicle position information, determine the target position that the vehicle **100** should go next, and generate a route from the current position of the vehicle **100** indicated in the acquired vehicle position information to the target position. The server **200** may generate a route to a target position which is between the current position and the target position or may generate a route to the target position. The server **200** may transmit the generated route to the vehicle **100**. The vehicle **100** may generate the travel control signal in such a way that the vehicle **100** travels along the route received from the server **200** and control the actuators **120** using the generated travel control signal.

[0133] (2) The server **200** may acquire vehicle position information and transmit the acquired vehicle position information to the vehicle **100**. The vehicle **100** may determine the target position that the vehicle **100** should go next, generate a route from the current position of the vehicle **100** indicated in the received vehicle position information to the target position, generate a travel control signal in such a way that the vehicle **100** travels along the generated route, and control the actuators **120** using the generated travel control signal.

[0134] (3) In the forms of the above (1) and (2), an internal sensor may be mounted on the vehicle **100**, and a result of detection output from the internal sensor may be used in at least one of the generation of the route or the generation of the travel control signal. The internal sensor is a sensor mounted on the vehicle **100**. The internal sensor may include, for example, a sensor that detects a motion state of the vehicle **100**, a sensor that detects an operation state of each part of the vehicle **100**, and a sensor that detects an environment near the vehicle **100**. Specifically, the internal sensor may include, for example, a camera, LiDAR, a millimeter wave radar, an ultrasonic sensor, a GPS sensor, an acceleration sensor, a gyro sensor or the like. For example, in the form of the above (1), the server **200** may acquire the result of the detection in the internal sensor and reflect the result of the detection in the internal sensor in the route when the route is generated. In the form of the above (1), the vehicle **100** may acquire the result of the detection in the internal sensor and reflect the result of the detection in the internal sensor in a travel control signal when the travel control signal is generated. In the form of the above (2), the vehicle **100** may acquire the result of the detection in the internal sensor, and reflect the result of the detection in the internal sensor in the route when the route is generated. In the form of the above (2), the vehicle **100** may acquire the result of the detection in the internal sensor, and reflect the result of the detection in the internal sensor in the travel control signal when the travel control signal is generated.

[0135] (YY3) In the travel control example 2, an internal sensor may be mounted on the vehicle **100***v* and the result of the detection output from the internal sensor may be used in at least one of the generation of the route or the generation of the travel control signal. For example, the vehicle **100***v* may acquire the result of the detection in the internal sensor, and reflect the result of the detection in the internal sensor in the route when the route is generated. The vehicle **100***v* may acquire the result of the detection in the internal sensor and reflect the result of the detection in the internal sensor in the travel control signal when the travel control signal is generated.

[0136] (YY4) In the travel control example 2, the vehicle **100***v* acquires the vehicle position information using the result of the detection by the sensor **300**. On the other hand, an internal sensor is mounted on the vehicle **100***v*, and the vehicle **100***v* may acquire vehicle position information using a result of detection in the internal sensor, determine the target position that the vehicle **100***v* should go next, generate a route from the current position of the vehicle **100***v* indicated in the acquired vehicle position information to the target position, generate a travel control signal for traveling along the generated route, and control the actuators **120** using the generated travel control signal. In this case, the vehicle **100***v* may travel without using the result of the detection by the sensor **300**. Note that the vehicle **100***v* may acquire a target arrival time or congestion information from the outside of the vehicle **100***v* and reflect the target arrival time or the congestion information in at least one of the route or the travel control signal. Further, all the functional configurations of the system **50***v* may be provided in the vehicle **100***v*. That is, the processing implemented by the system **50***v* in the present disclosure may be achieved by the vehicle **100***v* alone. For example, a leading vehicle **100***v* may transmit control instruction values to the following vehicle **100**.

[0137] (YY5) In the travel control example 1, the server **200** automatically generates a travel control signal to be transmitted to the vehicle **100**. On the other hand, the server **200** may generate the travel control signal to be transmitted to the vehicle **100** in accordance with an operation performed by an external operator located in the outside of the vehicle **100**. For example, the external operator may operate a manipulation apparatus including a display for displaying a captured image output from the sensor **300**, a steering, an accelerator pedal, and a brake pedal for remotely operating the vehicle **100**, and a communication apparatus for communicating with the server **200** by wired communication or wireless communication, and the server **200** may generate a travel control signal in accordance with the operation added to the manipulation apparatus.

[0138] (YY6) In each of the above travel control examples, it is sufficient that the vehicle **100** include a configuration capable of moving by unmanned driving, and the vehicle **100** may have, for example, a form of a platform including the configurations stated below. Specifically, it is sufficient that the vehicle **100** at least include the vehicle control apparatus **110** and the actuators **120** in order to exert three functions of “run”, “turn”, and “stop” by unmanned driving. In a case where the vehicle **100** externally acquires information for unmanned driving, the vehicle **100** may further include a communication apparatus **130**. That is, the vehicle **100** that can move by unmanned driving may not be provided with at least some of internal components such as a driving seat or a dashboard, at least some of the external components such as a bumper or a fender, or a body shell. In this case, before the vehicle **100** is shipped from the factory FC, the other components such as a body shell may be mounted on the vehicle **100**, or the other components such as the body shell may be mounted on the vehicle **100** after the vehicle **100** is shipped from the factory FC in a state in which the other components such as the body shell are not mounted on the vehicle **100**. Each of the components may be mounted thereon from a desired direction such as an upper side, a lower side, a front side, a rear side, a right side, or a left side of the vehicle **100**, mounted thereon from the same direction, or mounted thereon from different directions. In terms of the form of the platform, the position may be determined as in the vehicle **100** according to the first embodiment.

[0139] (YY7) The vehicle **100** may be manufactured by combining a plurality of modules. The module means a unit formed of a plurality of components grouped according to a part or a function of the vehicle **100**. For example, the platform of the vehicle **100** may be manufactured by combining a front module that forms a front part of the platform, a central module that forms a central part of the platform, and a rear module that forms a rear part of the platform with one another. Note that the number of modules that form the platform is not limited to three, and may be two or smaller or four or larger. Further, in addition to or in place of the components that form the platform, components of the vehicle **100** that form parts other than the platform may be formed in modules. Further, these modules may include any exterior components such as a bumper or a grill or any interior components such as seats and a console. Further, not only the vehicle **100** but also any form of mobile body may be manufactured by combining a plurality of modules. These modules may be manufactured, for example, by joining a plurality of components by welding, fixtures, or the like, or may be manufactured by integrally molding at least some of the components that form the module as one component by casting. A molding method of integrally forming one component, in particular, a relatively large-sized component is also called gigacasting or megacasting. For example, the above front module, central module, and rear module may be manufactured using gigacasting.
[0140] (YY8) Transporting a vehicle **100** using traveling of the vehicle **100** by unmanned driving is also referred to as “self-propelled transportation”. Further, a configuration for achieving self-propelled transportation is referred to as a “vehicle remote control autonomous travel transportation system”. Further, a production method for producing vehicles **100** using self-propelled transportation is also referred to as “self-propelled production”. In the self-propelled production, for example, in a factory FC that manufactures the vehicles **100**, a part of the transportation of the vehicle **100** is achieved by self-propelled transportation.
[0141] (YY9) In each of the above travel control examples, some or all of the functions and processing implemented in the form of software may be implemented in the form of hardware. Further, some or all of the functions and processing implemented in the form of hardware may be implemented in the form of software. For example, various types of circuits such as an integrated circuit or a discrete circuit may be used as hardware for implementing various types of functions in each of the above embodiments.
[0142] Further, some or all of the processing performed in the vehicle **100**, the server **200**, the sensor **300**, the robot **600**, and so on described above may be implemented as a computer program.
[0143] A (The) program can be stored and provided to a computer using any type of non-transitory computer readable media. Non-transitory computer readable media include any type of tangible storage media. Examples of non-transitory computer readable media include magnetic storage media (such as floppy disks, magnetic tapes, hard disk drives, etc.), optical magnetic storage media (e.g. magneto-optical disks), CD-ROM (compact disc read only memory), CD-R (compact disc recordable), CD-R/W (compact disc rewritable), and semiconductor memories (such as mask ROM, PROM (programmable ROM), EPROM (erasable PROM), flash ROM, RAM (random access memory), etc.). The program may be provided to a computer using any type of transitory computer readable media. Examples of transitory computer readable media include electric signals, optical signals, and electromagnetic waves. Transitory computer readable media can provide the program to a computer via a wired communication line (e.g. electric wires, and optical fibers) or a wireless communication line.
[0144] From the disclosure thus described, it will be obvious that the embodiments of the disclosure may be varied in many ways. Such variations are not to be regarded as a departure from the spirit and scope of the disclosure, and all such modifications as would be obvious to one skilled in the art are intended for inclusion within the scope of the following claims.

Claims

1. A vehicle manufacturing system for performing control in such a way that a plurality of vehicles forming a platoon travel in a manufacturing process or a transporting process, the vehicle manufacturing system comprising: a computing unit for computing a control instruction value for controlling a speed of the vehicle; a transmitter configured to transmit the control instruction value to vehicles included in a group including two or more of the vehicles included in the platoon, the control instruction value being common among the vehicles included in the group; a speed control unit for controlling the speed of the vehicle in accordance with the control instruction value; a sensor provided to detect an inter-vehicle distance, which is a distance between the vehicles; and a group adjustment unit for changing the vehicles included in the group in accordance with the inter-vehicle distance.
2. The vehicle manufacturing system according to claim 1, wherein an approaching vehicle whose inter-vehicle distance between it and a front vehicle has become smaller than a first threshold is detected, and the approaching vehicle is separated from a group including the front vehicle which is traveling ahead of the approaching vehicle.
3. The vehicle manufacturing system according to claim 1, wherein a delayed vehicle whose inter-vehicle distance between it and the front vehicle has become greater than a second threshold is detected, and the front vehicle which is traveling ahead of the delayed vehicle and the delayed vehicle are categorized in groups different from each other.
4. The vehicle manufacturing system according to claim 1, wherein in a case where it is detected that a variation in the inter-vehicle distance has become greater than a third threshold, the computing unit individually computes the control instruction values for the respective vehicles, and the transmitter transmits the control instruction values to the respective vehicles.
5. The vehicle manufacturing system according to claim 1, wherein the transmitter transmits an individual control instruction value to the vehicle that has reached a predicted part where it is expected that the inter-vehicle distance will vary.
6. The vehicle manufacturing system according to claim 1, comprising: a camera configured to capture an image of any one of the vehicles in the platoon, wherein the computing unit computes the control instruction value based on a result of capturing the image in the camera.
7. A vehicle manufacturing method for performing control in such a way that a plurality of vehicles forming a platoon travel in a manufacturing process or a transporting process, the vehicle manufacturing method comprising: computing a control instruction value for controlling a speed of the vehicle; detecting, based on a result of detection in a sensor, an inter-vehicle distance, which is a distance between one of the plurality of vehicles and another adjacent one of the plurality of vehicles; transmitting, by a transmitter, the control instruction value to vehicles included in a group including two or more of the vehicles included in the platoon, the control instruction value being common among the vehicles included in the group; controlling the speed of the vehicle in accordance with the control instruction value; and changing the vehicles included in the group in accordance with the inter-vehicle distance.
8. The vehicle manufacturing method according to claim 7, comprising: detecting an approaching vehicle whose inter-vehicle distance between it and a front vehicle has become smaller than a first threshold; and separating the approaching vehicle from a group including the front vehicle which is traveling ahead of the approaching vehicle.
9. The vehicle manufacturing method according to claim 7, comprising: detecting a delayed vehicle whose inter-vehicle distance between it and the front vehicle has become greater than a second threshold, and categorizing the front vehicle which is traveling ahead of the delayed vehicle and the delayed vehicle in groups different from each other.
10. The vehicle manufacturing method according to claim 7, wherein in a case where it is detected

that a variation in the inter-vehicle distance has become greater than a third threshold, a computing unit individually computes the control instruction values for the respective vehicles, and the transmitter transmits the control instruction values to the respective vehicles.

11. The vehicle manufacturing method according to claim 7, wherein the transmitter transmits an individual control instruction value to the vehicle that has reached a predicted part where it is expected that the inter-vehicle distance will vary.

12. The vehicle manufacturing method according to claim 7, comprising computing the control instruction value based on a result of capturing an image in a camera configured to capture an image of any one of the vehicles in the platoon.